

E

OPERATIONS WITH PLANE VECTORS

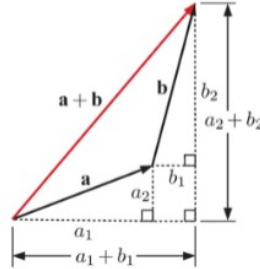
ALGEBRAIC VECTOR ADDITION

Consider adding vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$.

Notice that:

- the horizontal step for $\mathbf{a} + \mathbf{b}$ is $a_1 + b_1$
- the vertical step for $\mathbf{a} + \mathbf{b}$ is $a_2 + b_2$.

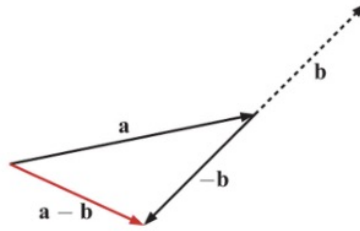
If $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ then $\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \end{pmatrix}$.



$$\vec{a} + \vec{b} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \end{pmatrix}$$

Suppose $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$.

$$\begin{aligned}\therefore \mathbf{a} - \mathbf{b} &= \mathbf{a} + (-\mathbf{b}) \\ &= \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} -b_1 \\ -b_2 \end{pmatrix} \\ &= \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \end{pmatrix}\end{aligned}$$



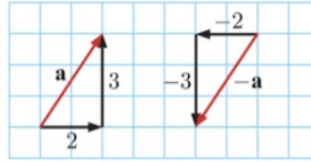
If $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$, then $\mathbf{a} - \mathbf{b} = \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \end{pmatrix}$.

ALGEBRAIC NEGATIVE VECTORS

In the diagram we see the vector $\mathbf{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

and its negative $-\mathbf{a} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$.

$$\text{If } \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \text{ then } -\mathbf{a} = \begin{pmatrix} -a_1 \\ -a_2 \end{pmatrix}.$$

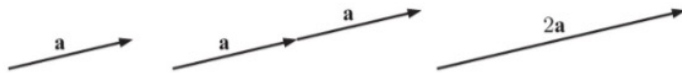


ALGEBRAIC VECTOR SUBTRACTION

To subtract one vector from another, we simply **add its negative**.

ALGEBRAIC SCALAR MULTIPLICATION

We have already seen a geometric approach for integer scalar multiplication:



Consider $\mathbf{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$. $2\mathbf{a} = \mathbf{a} + \mathbf{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \times 1 \\ 2 \times 3 \end{pmatrix}$
 $3\mathbf{a} = \mathbf{a} + \mathbf{a} + \mathbf{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 9 \end{pmatrix} = \begin{pmatrix} 3 \times 1 \\ 3 \times 3 \end{pmatrix}$

If k is any scalar and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, then $k\mathbf{v} = \begin{pmatrix} kv_1 \\ kv_2 \end{pmatrix}$.

Notice that:

- $(-1)\mathbf{v} = \begin{pmatrix} (-1)v_1 \\ (-1)v_2 \end{pmatrix} = \begin{pmatrix} -v_1 \\ -v_2 \end{pmatrix} = -\mathbf{v}$
- $(0)\mathbf{v} = \begin{pmatrix} (0)v_1 \\ (0)v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \mathbf{0}$

$\vec{a} \rightarrow 2\vec{a}, 3\vec{a}$
in general $k\vec{a}$

$\therefore \vec{a} \parallel k\vec{a}$, $k > 0 \rightarrow$ They are in same direction

$k < 0 \rightarrow$ They are in opposite direction.

EXERCISE 12E

1 If $\mathbf{a} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} -2 \\ -5 \end{pmatrix}$ find:

a $\mathbf{a} + \mathbf{b}$

b $\mathbf{b} + \mathbf{a}$

c $\mathbf{b} + \mathbf{c}$

d $\mathbf{c} + \mathbf{b}$

e $\mathbf{a} + \mathbf{c}$

f $\mathbf{c} + \mathbf{a}$

g $\mathbf{a} + \mathbf{a}$

h $\mathbf{b} + \mathbf{a} + \mathbf{c}$

$$(1) \quad \vec{a} + \vec{b} = \begin{pmatrix} -3 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$$

2 Given $\mathbf{p} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -1 \\ -5 \end{pmatrix}$, and $\mathbf{r} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ find:

12E

a $\mathbf{p} - \mathbf{q}$

b $\mathbf{q} - \mathbf{r}$

c $\mathbf{p} + \mathbf{q} - \mathbf{r}$

d $\mathbf{p} - \mathbf{q} - \mathbf{r}$

e $\mathbf{q} - \mathbf{r} - \mathbf{p}$

f $\mathbf{r} + \mathbf{q} - \mathbf{p}$

3 Let $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.
12E

a Use vector addition to show that $\mathbf{a} + \mathbf{0} = \mathbf{a}$.

b Use vector subtraction to show that $\mathbf{a} - \mathbf{a} = \mathbf{0}$.

$$(a) \quad \vec{a} + \vec{0} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} a_1 + 0 \\ a_2 + 0 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \vec{a}$$

$$(b) \quad \vec{a} - \vec{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} -a_1 \\ -a_2 \end{pmatrix} = \begin{pmatrix} a_1 - a_1 \\ a_2 - a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \vec{0}$$

4 For $\mathbf{p} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$, and $\mathbf{r} = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$ find:

12E

a $-3\mathbf{p}$

b $\frac{1}{2}\mathbf{q}$

c $2\mathbf{p} + \mathbf{q}$

d $\mathbf{p} - 2\mathbf{q}$

e $\mathbf{p} - \frac{1}{2}\mathbf{r}$

f $2\mathbf{p} + 3\mathbf{r}$

g $2\mathbf{q} - 3\mathbf{r}$

h $2\mathbf{p} - \mathbf{q} + \frac{1}{3}\mathbf{r}$

$$(a) -3\vec{p} = -3 \begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} -3 \\ -15 \end{pmatrix}$$

5 Consider $\mathbf{p} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

RE

a Find geometrically:

i $\mathbf{p} + \mathbf{p} + \mathbf{q} + \mathbf{q} + \mathbf{q}$

ii $\mathbf{p} + \mathbf{q} + \mathbf{p} + \mathbf{q} + \mathbf{q}$

iii $\mathbf{q} + \mathbf{p} + \mathbf{q} + \mathbf{p} + \mathbf{q}$

b Comment on your results in **a**.

6 Find k such that:

12E a $\begin{pmatrix} 2 \\ -5 \end{pmatrix} + k \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 20 \\ -11 \end{pmatrix}$

b $\frac{1}{3} \begin{pmatrix} -1 \\ k \end{pmatrix} + \frac{4}{3} \begin{pmatrix} 7 \\ k+1 \end{pmatrix} = \begin{pmatrix} 9 \\ -1 \end{pmatrix}$

$$(a) \begin{pmatrix} 2+3k \\ -5-k \end{pmatrix} = \begin{pmatrix} 20 \\ -11 \end{pmatrix}$$

$$\therefore 2+3k=20$$

$$3k=18$$

$$\boxed{k=6}$$

$$-5-k=-11$$

$$-k=-11+5=-6$$

$$\boxed{k=6}$$

✓
7 If $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{s} = -\mathbf{i} + 4\mathbf{j}$, find:
12E a $|\mathbf{r}|$ b $|\mathbf{s}|$ ✓ c $|\mathbf{r} + \mathbf{s}|$ d $|\mathbf{r} - \mathbf{s}|$
e $|\mathbf{s} - 2\mathbf{r}|$ f $|2\mathbf{r}|$ g $|\mathbf{r} + 2\mathbf{s}|$ h $|2\mathbf{r} - \mathbf{s}|$

$$(a) |\vec{r}| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

$$(b) |\vec{s}| = \sqrt{(-1)^2 + (4)^2} = \sqrt{1 + 16} = \sqrt{17}$$

$$(c) \vec{r} + \vec{s} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

$$\therefore |\vec{r} + \vec{s}| = \sqrt{1^2 + 7^2} = \underline{\underline{\sqrt{50}}}$$

- 8 a If $\mathbf{p} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, find:
- 12E i $|\mathbf{p}|$ ii $|2\mathbf{p}|$ iii $|-2\mathbf{p}|$ iv $|3\mathbf{p}|$ v $|-3\mathbf{p}|$
- b By letting $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, prove that $|k\mathbf{v}| = |k||\mathbf{v}|$.

$$|k\vec{v}| = \begin{pmatrix} |kv_1| \\ |kv_2| \end{pmatrix}$$

$$|k\vec{v}| = \sqrt{|k|^2 v_1^2 + |k|^2 v_2^2} = |k| \sqrt{v_1^2 + v_2^2} = |k| |\underline{\underline{\vec{v}}}|$$

9 Let $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.

12E

Show by equating components, that if $k\mathbf{x} = \mathbf{a}$ then $\mathbf{x} = \frac{1}{k}\mathbf{a}$.

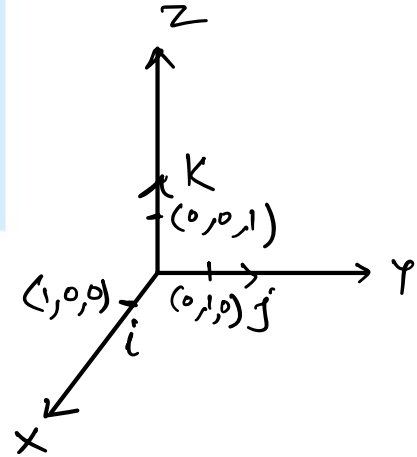
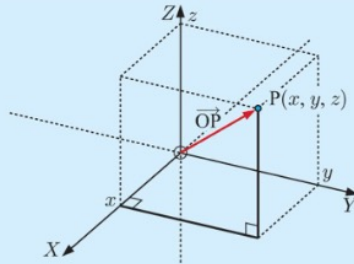
In **Chapter 10** of the **Core Topics HL** book, we saw that we can specify a point in 3-dimensional space using 3 mutually perpendicular axes called the X , Y , and Z axes.

In 3-dimensional space, the **position vector** of $P(x, y, z)$

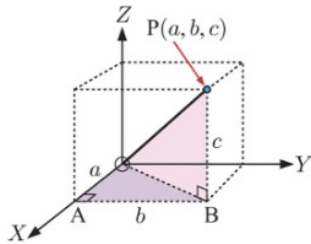
$$\text{is } \vec{OP} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

$$\text{where } \mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \text{ and } \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

are the **base unit vectors** in the X , Y , and Z directions respectively.



THE MAGNITUDE OF A VECTOR



Triangle OAB is right angled at A

$$\therefore OB^2 = a^2 + b^2 \quad \dots (1) \quad \{\text{Pythagoras}\}$$

Triangle OBP is right angled at B

$$\therefore OP^2 = OB^2 + c^2 \quad \{\text{Pythagoras}\}$$

$$\therefore OP^2 = a^2 + b^2 + c^2 \quad \{\text{using (1)}\}$$

$$\therefore OP = \sqrt{a^2 + b^2 + c^2}$$

The magnitude or length of the vector $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ is $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$.

EXERCISE 12F

1 Consider the point $T(3, -1, 4)$.

- a Draw a diagram to locate the position of T in space.
- c Find the distance from O to T.

b Find $\vec{OT}$.

$$(b) \quad \vec{OT} = \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix}$$

$$(c) \quad |\vec{OT}| = \sqrt{3^2 + 1^2 + 4^2} \\ = \underline{\underline{\sqrt{26}}}$$

position vector
any vector originating
from origin is called
position.

2 For each point P, write $\vec{OP}$ in component form and unit vector form:

12F a P(2, 0, 4)

b P(-3, 1, 2)

c P(0, -2, 0)

$$(a) \vec{OP} = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} \quad \& \quad \vec{OP} = 2i + 0j + 4k = \underline{2i + 4k}$$

3 Find the length of:

12F

a $\begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}$

b $\begin{pmatrix} -4 \\ 0 \\ 3 \end{pmatrix}$

c $\begin{pmatrix} 4 \\ 1 \\ -2 \end{pmatrix}$

d $\begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}$

$$(d) \left| \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix} \right| = \sqrt{4+4+1} = \sqrt{9} = \underline{\underline{3}}$$

4 Illustrate P and find its distance from the origin O:

12F a $P(0, 0, -3)$

b $P(0, -1, 2)$

c $P(3, 1, 4)$

d $P(-1, -2, 3)$

(a) $P(0, 0, -3)$ → P lies on -ve -z-axis → $|\vec{OP}| = 3$

(b) $P(0, -1, 2)$ → P is on yz-plane.

8-Octants Points in Spaces

1 • $OXYZ$ (a, b, c)

2 • $OXYZ'$ → (a, b, -c)

3 • $OX'YZ$ → (a, -b, c)

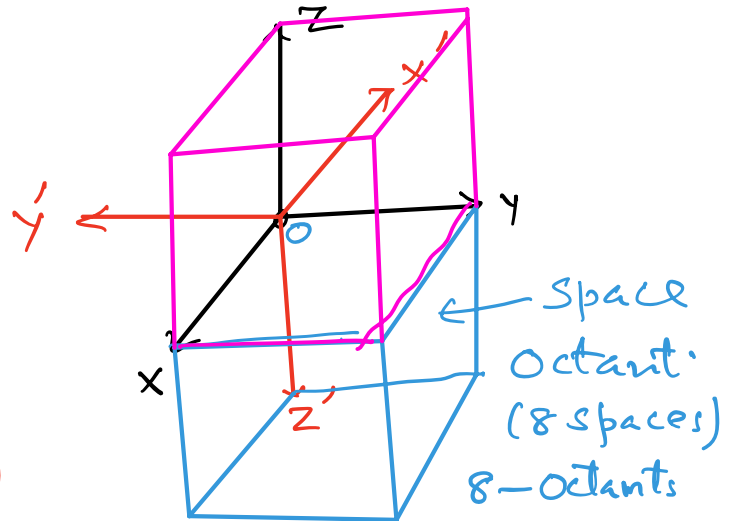
4 • $OX'YZ$ → (-a, b, c)

5 • $OX'YZ'$ → (-a, b, -c)

6 • $OX'YZ$ → (-a, b, c)

7 • $OX'YZ$ → (-a, -b, c)

8 • $OX'YZ'$ → (-a, -b, -c)



Planes: (1) $XY\text{-plane} \rightarrow (a, b, 0)$ (a, b, c)

(2) $YZ\text{-plane} \rightarrow (0, b, c)$

(3) $ZX\text{-plane} \rightarrow (a, 0, c)$

(4) $X'Y\text{-plane} \rightarrow (-a, b, 0)$

- - - - -

5 Write each vector in component form, and find its length:

12F a $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$

b $\mathbf{b} = -3\mathbf{i} + 4\mathbf{k}$

c $\mathbf{c} = 2\mathbf{j} - 3\mathbf{k}$

$$(4) \quad \vec{a} = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$$

$$|\vec{a}| = \sqrt{2^2 + 1^2 + 4^2} = \sqrt{21}$$

6 Find a , b , and c if:

12F

a $\begin{pmatrix} a-4 \\ b-3 \\ c+2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}$

b $\begin{pmatrix} a-5 \\ b-2 \\ c+3 \end{pmatrix} = \begin{pmatrix} 3-a \\ 2-b \\ 5-c \end{pmatrix}$

c $\begin{pmatrix} 3a+4 \\ b-c \\ 4-5b \end{pmatrix} = \begin{pmatrix} 12-a \\ a-4 \\ 5a-22 \end{pmatrix}$

(a) $a-4=1 \Rightarrow a=5$

$b-3=3 \Rightarrow b=6$

$c+2=-4 \rightarrow c=-6$

7 Which of the following are unit vectors?

12F

a $\begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$

b $\begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$

c $\begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{pmatrix}$

d $\begin{pmatrix} \frac{1}{3\sqrt{3}} \\ \frac{5}{3\sqrt{3}} \\ -\frac{1}{3\sqrt{3}} \end{pmatrix}$

A unit vector has length 1.

$$(a) \left| \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \right| = \sqrt{1^2} = 1 \checkmark$$

$$(b) \left| \begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \right| = \sqrt{0 + \frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{2}{4}} = \frac{\sqrt{2}}{2} \times$$

$$(c) \left| \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{pmatrix} \right| = \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2} = \sqrt{\frac{2^2 + 1^2 + 2^2}{3}} = \frac{3}{3} = 1 \checkmark$$

$$(d) \left| \begin{pmatrix} \frac{1}{3\sqrt{3}} \\ \frac{5}{3\sqrt{3}} \\ -\frac{1}{3\sqrt{3}} \end{pmatrix} \right| = \sqrt{\left(\frac{1}{3\sqrt{3}}\right)^2 + \left(\frac{5}{3\sqrt{3}}\right)^2 + \left(\frac{1}{3\sqrt{3}}\right)^2}$$

$$= \sqrt{\frac{1}{27} + \frac{25}{27} + \frac{1}{27}} = \sqrt{\frac{27}{27}} = 1 \checkmark$$

8 Find k given the unit vector:

12F

a $\begin{pmatrix} -\frac{1}{2} \\ k \\ \frac{1}{4} \end{pmatrix}$

b $\begin{pmatrix} k \\ \frac{2}{3} \\ -\frac{1}{3} \end{pmatrix}$

c $\begin{pmatrix} k \\ k \\ k \end{pmatrix}$

d $\begin{pmatrix} \frac{1}{k} \\ 2 \\ \frac{3}{k} \end{pmatrix}$

$$(a) \left| \begin{pmatrix} -\frac{1}{2} \\ k \\ \frac{1}{4} \end{pmatrix} \right| = 1$$

$$\sqrt{\frac{1}{4} + k^2 + \frac{1}{16}} = 1$$

$$\frac{1}{4} + k^2 + \frac{1}{16} = 1$$

$$k^2 = 1 - \frac{1}{4} - \frac{1}{16} = \frac{16-4-1}{16} = \frac{11}{16}$$

$$\therefore k = \pm \frac{\sqrt{11}}{4}$$

$$(c) \left| \begin{pmatrix} k \\ k \\ k \end{pmatrix} \right| = 1$$

$$\sqrt{3k^2} = 1$$

$$3k^2 = 1$$

$$k^2 = \frac{1}{3}$$

$$k = \pm \frac{1}{\sqrt{3}}$$

9 Find the possible values of m , given that:

12F a $\begin{pmatrix} -2 \\ m \\ 1 \end{pmatrix}$ has length $\sqrt{14}$ units

b $\begin{pmatrix} 4 \\ -3 \\ m \end{pmatrix}$ has length 6 units.

(a)

$$\left| \begin{pmatrix} -2 \\ m \\ 1 \end{pmatrix} \right| = \sqrt{14}$$

$$\sqrt{4 + m^2 + 1} = \sqrt{14}$$

$$4 + m^2 + 1 = 14$$

$$m^2 = 14 - 5$$

$$m^2 = 9$$

$$m = \pm 3$$

10 Suppose $\left| \begin{pmatrix} a \\ b \end{pmatrix} \right| = 5$ units and $\left| \begin{pmatrix} b \\ c \end{pmatrix} \right| = 3$ units. Find the range of possible values of $\left| \begin{pmatrix} a \\ b \\ c \end{pmatrix} \right|$.

12F

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \sqrt{a^2 + b^2} = 5 & \sqrt{b^2 + c^2} = 3 & \sqrt{a^2 + b^2 + c^2} \\ a^2 + b^2 = 25 & b^2 + c^2 = 9 & \\ \text{---(i)} & \text{---(ii)} & \end{array}$$

$$a^2 + \underline{2b^2} + c^2 = 25 + 9 = 34$$

$$\Rightarrow 0 < \left| \begin{pmatrix} a \\ b \\ c \end{pmatrix} \right| < 34$$

$$\left| \begin{pmatrix} a \\ b \end{pmatrix} \right| = 5$$

$$a = \pm 3, b = \pm 4$$

$$\sqrt{a^2 + b^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

$$\begin{array}{l} \sqrt{b^2 + c^2} = 3 \quad \begin{array}{r} 81 \\ -16 \\ \hline 65 \end{array} \\ b = \pm 4, c = \pm \sqrt{65} \\ \sqrt{16 + 65} = \sqrt{81} = 9 \end{array}$$

$$\left| \begin{pmatrix} a \\ b \\ c \end{pmatrix} \right| = \sqrt{9 + 16 + 65} = \sqrt{25 + 65} = \underline{\underline{\sqrt{90}}}$$

For 3-dimensional vectors, it is difficult to illustrate operations. Instead, we perform operations using component form.

The rules for algebra with vectors readily extend from 2 dimensions to 3 dimensions:

If $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ then:

- $-\mathbf{a} = \begin{pmatrix} -a_1 \\ -a_2 \\ -a_3 \end{pmatrix}$

- $\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}$

- $\mathbf{a} - \mathbf{b} = \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \\ a_3 - b_3 \end{pmatrix}$

- $k\mathbf{a} = \begin{pmatrix} ka_1 \\ ka_2 \\ ka_3 \end{pmatrix}$ for any scalar k .

EXERCISE 12G

1 Let $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}$. Find:

a $\mathbf{a} + \mathbf{b}$

d $\mathbf{c} - \frac{1}{2}\mathbf{a}$

b $\mathbf{a} - \mathbf{b}$

e $\mathbf{a} - \mathbf{b} - \mathbf{c}$

c $\mathbf{b} + 2\mathbf{c}$

f $2\mathbf{b} - \mathbf{c} + \mathbf{a}$

2 Let $\mathbf{a} = \mathbf{i} + 3\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} + \mathbf{j} + \mathbf{k}$. Find:

129 **a** $|\mathbf{a}|$

b $|\mathbf{b}|$

c $2|\mathbf{a}|$

d $|2\mathbf{a}|$

e $-3|\mathbf{b}|$

f $|-3\mathbf{b}|$

g $|\mathbf{a} + \mathbf{b}|$

h $|\mathbf{a} - \mathbf{b}|$

129 **3** Let $\mathbf{a} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} -2 \\ 2 \\ 4 \end{pmatrix}$. Find:

a $2\mathbf{a} - \mathbf{c}$

b $3\mathbf{b} + \frac{1}{2}\mathbf{c}$

c $-\mathbf{a} + 3\mathbf{b}$

d $\mathbf{a} - 5\mathbf{b} + 4\mathbf{c}$

e $|\mathbf{b} + \mathbf{c}|$

f $|\mathbf{a} - \mathbf{c}|$

g $|\mathbf{a}| |\mathbf{b}|$

h $\frac{1}{|\mathbf{a}|} \mathbf{a}$

4 Find scalars a , b , and c such that:

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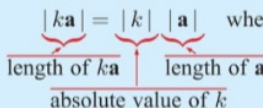
a $2 \begin{pmatrix} 1 \\ 0 \\ 3a \end{pmatrix} = \begin{pmatrix} b \\ c-1 \\ 2 \end{pmatrix}$

c $a \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + b \begin{pmatrix} 1 \\ 7 \\ 2 \end{pmatrix} = \begin{pmatrix} 7 \\ c \\ 2 \end{pmatrix}$

b $a \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} + c \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ 3 \end{pmatrix}$

d $2 \begin{pmatrix} a \\ 1 \\ -2 \end{pmatrix} + \begin{pmatrix} b \\ c \\ 4 \end{pmatrix} = \begin{pmatrix} c \\ -c \\ ab \end{pmatrix}$

From the **Investigation** you should have found that for vectors **a, b, c** and scalar $k \in \mathbb{R}$:

- $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ {commutative property}
- $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$ {associative property}
- $\mathbf{a} + \mathbf{0} = \mathbf{0} + \mathbf{a} = \mathbf{a}$ {additive identity}
- $\mathbf{a} + (-\mathbf{a}) = (-\mathbf{a}) + \mathbf{a} = \mathbf{0}$ {additive inverse}
- $k(\mathbf{a} + \mathbf{b}) = k\mathbf{a} + k\mathbf{b}$ {distributive property}
- $|k\mathbf{a}| = |k| |\mathbf{a}|$ where $k\mathbf{a}$ is parallel to $\mathbf{a}$


The rules for solving vector equations are similar to those for solving real number equations. However, there is no such thing as dividing a vector by a scalar. Instead, we multiply by the reciprocal scalar.

For example, if $2\mathbf{x} = \mathbf{a}$ then $\mathbf{x} = \frac{1}{2}\mathbf{a}$ and *not* $\frac{\mathbf{a}}{2}$. $\frac{\mathbf{a}}{2}$ has no meaning in vector algebra.

Two useful rules are:

- If $\mathbf{x} + \mathbf{a} = \mathbf{b}$ then $\mathbf{x} = \mathbf{b} - \mathbf{a}$.
- If $k\mathbf{x} = \mathbf{a}$, $k \neq 0$, then $\mathbf{x} = \frac{1}{k}\mathbf{a}$.

EXERCISE 12H

1 Solve the following vector equations for $\mathbf{x}$:

a $2\mathbf{x} = \mathbf{q}$

b $\frac{1}{2}\mathbf{x} = \mathbf{n}$

c $-3\mathbf{x} = \mathbf{p}$

d $\mathbf{q} + 2\mathbf{x} = \mathbf{r}$

e $4\mathbf{s} - 5\mathbf{x} = \mathbf{t}$

f $4\mathbf{m} - \frac{1}{3}\mathbf{x} = \mathbf{n}$

End of Lesson — 8/10/2026